

Beyond 550 Tb/s S+C+L-band Bidirectional Transmission over 10.9-km Anti-Resonant Hollow-Core Fiber

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Abstract: We demonstrate a same-wavelength bidirectional transmission over a 10.9-km AR-HCF with 393×2 channels and 50-GHz spaced grid. The GMI-estimated net rate of 550.97 Tb/s indicates a record capacity for any single-core single-mode fiber. © 2026 The Author(s)

1. Introduction

Optical fibre communication is the cornerstone of the internet, which carries more than 90% of the global data traffic [1]. To meet the ever-increasing demand for data transmission capacity, various techniques have been proposed, including multi-band wavelength-division multiplexing (WDM), space-division multiplexing (SDM), probabilistic constellation shaping, nonlinear compensation, and deep learning, among others [2, 3]. In [4], a 402-Tb/s generalized mutual information (GMI) data-rate transmission was demonstrated over 50-km standard single-mode fibre (SSMF) by covering the full transmission bands (O/E/S/C/L/U). It represents the highest data rate reported in a single-core single-mode silica-core fibre. However, the fundamental limitations of silica-core-based SSMF prevent the rate from further improvement, which may lead to a predictable capacity crunch ultimately [5].

Anti-resonant hollow-core fibre (AR-HCF) is regarded as a brand-new light processing platform to break through the limitations of silica-core fibre owing to the unique merits, such as low transmission loss (<0.11 dB/km in 2024), low latency (about 30% lower latency than silica-core SSMF), low nonlinearity (~3 orders of magnitude lower than silica), low Rayleigh backscattering coefficient, low chromatic dispersion, and ultra-wide bandwidth [6-8]. Recently, several high-capacity transmission experiments about AR-HCF have been reported [9-13]. In our previous works, we demonstrated a record 502.6 Tb/s bidirectional transmission across the S+C+L bands over 11.04-km AR-HCF [14, 15]. Nevertheless, the full potential of AR-HCF for ultra-high-capacity transmission has not yet been fully exploited.

In this paper, we report a record capacity by using dense WDM coherent transmission with 50-GHz channel spacing using a self-developed low-loss AR-HCF. The fabricated AR-HCF is a nested anti-resonant nodeless fibre (NANF) with <0.2-dB/km (0.69-dB/km in [14, 15]) propagation loss at 1550 nm. By optimizing the number of equalizer taps, the system performance of the wavelength channel with H₂O or CO₂ absorptions is improved significantly. Besides, the nonlinearity induced by photoelectric devices and fibre is partially compensated with the help of machine learning (ML) technique. By leveraging five dimensions (amplitude, phase, polarization, wavelength, and direction) to carry information, we achieve 550.97-Tb/s net rate covering S+C+L band from 1466 nm to 1626 nm. To the best of our knowledge, we achieve the highest transmission capacity for any single-core single-mode fibre.

2. Experimental setup and DSP

The experimental setup for the same-wavelength bidirectional S+C+L-band transmission based on NANF is shown in Fig. 1(a). A sliding three-channel test band with a 50-GHz spaced grid is generated by using three independent tunable lasers. The positive outputs of a 4-channel 128-GSa/s arbitrary waveform generator (AWG) drive a dual-polarization in-phase and quadrature (DP-IQ) modulator to generate a 49-GBaud probabilistic constellation shaped 256-ary quadrature amplitude modulation (PCS-256QAM) signal for the channel under test. In parallel, the four negative outputs of the AWG drive another DP-IQ modulator to produce the PCS-256QAM signals for the neighbor

channels, thereby emulating the inter-channel crosstalk in a WDM system. The test band is split into two parts by a 50:50 coupler to emulate two transmitters and then combined with a dummy band through a 10:90 coupler. In the dummy band, the wavelength-selective switches (WSS) are employed to flatten the spectra and carve a sliding 150-GHz notch to accommodate the test band. We use two S/C/L WDM couplers to combine the S+C+L-band amplified spontaneous emission (ASE) sources.

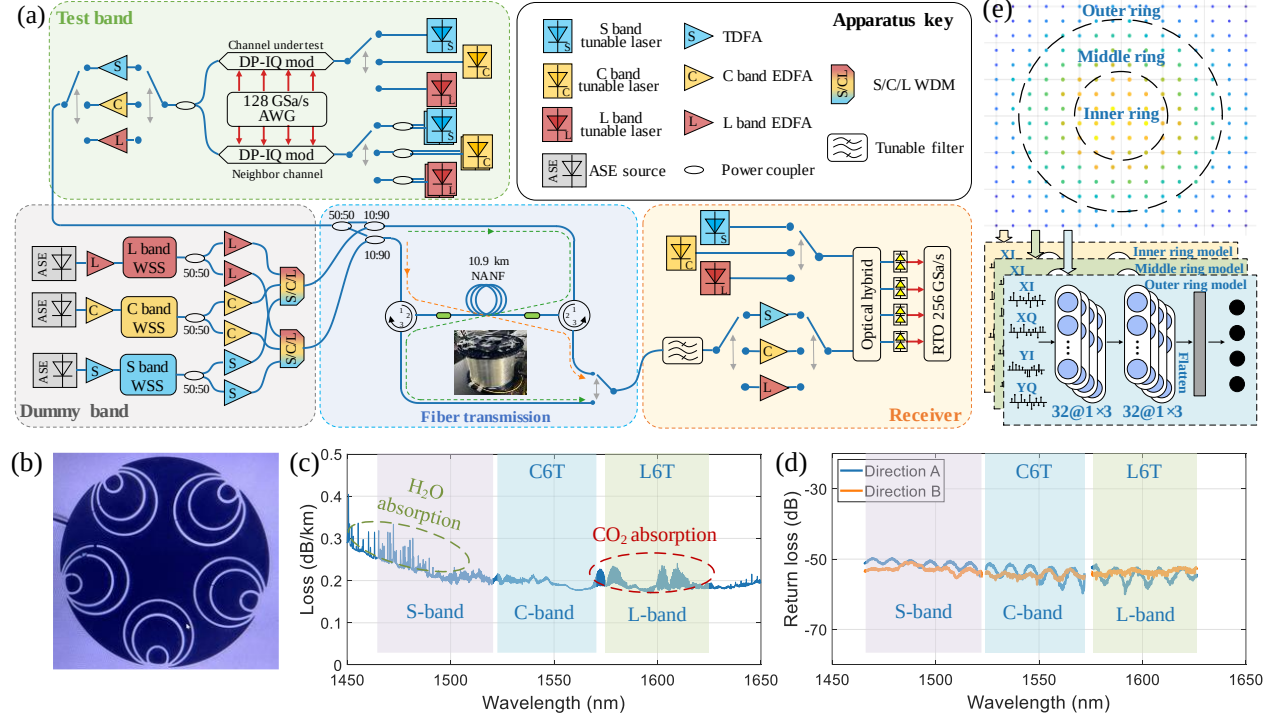


Fig. 1. (a) Experimental setup for same-wavelength bidirectional S+C+L-band transmission based on NANF. (b) Cross-section microstructure. (c) Propagation loss characteristic. (d) Return loss of the NANF. (e) Structure of the proposed RW-NNE.

Two circulators placed at both ends of the NANF are employed to enable the realization of the bidirectional transmission in a single-core fiber. Fig. 1(b) shows the cross-section microstructure of the self-developed NANF with a length of 10.9 km. The fabricated fiber adopts a triple-layer nested structure composed of five ring-shaped glass tubes. Fig. 1(c) shows that the measured propagation loss is < 0.2 dB/km at 1550 nm. Strong absorption peaks caused by water vapor and CO₂ can be observed, which increases the attenuation significantly. The measured return losses are lower than -50 dB in both directions across the full S+C+L bands, as depicted in Fig. 1(d), which is essential for achieving stable bidirectional transmission in a single-core fiber.

At the receiver side, we use an optical switch to select the output from either transmission direction. The signal of the channel under test is extracted by employing a tunable filter, which is subsequently detected by a coherent receiver and captured using a 59-GHz real-time oscilloscope (RTO) operating at 256 GSa/s. Finally, offline digital signal processing (DSP) is applied for impairment compensation and signal demodulation.

In the receiver DSP, a conventional feed-forward equalizer (FFE) based on the least mean square algorithm is used to compensate for the transmission impairments. Moreover, ML technique is utilized to compensate for the nonlinearity of the transmission link and transceivers after the FFE. The Ring-Wise Neural Network equalizer (RW-NNE) can mitigate the performance degradation caused by unevenly distributed training samples by partitioning constellation points into two ring regions for zone equalization of PCS-64QAM signals [16]. In the extended RW-NNE framework proposed herein, we employ supervised learning to train multiple sub-neural networks, as shown in Fig. 1(e). Specifically, PCS-256QAM signals are further partitioned into inner, middle, and outer rings based on constellation point amplitude characteristics, with overlapping regions established between adjacent rings. For each ring, a distinct convolutional neural network is constructed to learn the mapping relationship of the partitioned signal. Network weights are iteratively updated using the mean squared error (MSE) loss function. Signal samples from both the inner and outer edges of the demodulation boundary are simultaneously employed to train adjacent sub-networks, thus enhancing the robustness of boundary determination and mitigating allocation errors.

3. Experimental results

Fig. 2(a) illustrates the optical spectra of the bidirectional transmission, covering the S-band (1466 to 1522 nm), C-band (1524 to 1572 nm), and L-band (1575 to 1626 nm). The system comprises 153, 120, and 120 wavelength channels in the S, C, and L bands, respectively, with a 50-GHz spaced grid, resulting in 393×2 channels for both transmission directions. As shown in the figure, absorption peaks caused by H₂O and CO₂ after 10.9-km NANF transmission can be clearly identified, leading to noticeable signal degradation. Fig. 2(b) shows the GMI-estimated data rate as a function of the number of taps. In the presence of absorption peaks, increasing the number of taps improves system performance significantly. As illustrated in Fig. 2(c), up to 201 taps are employed, while only 61 taps are sufficient in the absence of absorption peaks. Fig. 2(d) shows the GMI-estimated net data rate after 10.9-km NANF. An entropy of 7.9 is applied uniformly across the S, C, and L bands. By leveraging the RW-NNE, the total net throughput is enhanced from 540.66 to 550.97 Tb/s, corresponding to 214.34, 170.6, and 166.03 Tb/s in the S, C, and L bands, respectively. This result verifies the effectiveness of RW-NNE in mitigating nonlinear distortions in NANF-based transmission. Compared with our previous work [14], the achieved capacity represents an ~10% improvement, enabled by the combination of lower-loss NANF and advanced DSP algorithms. The inset (i) presents the measured signal spectra with H₂O absorption. The insets (ii-v) show the recovered constellations at 1533.66 nm.

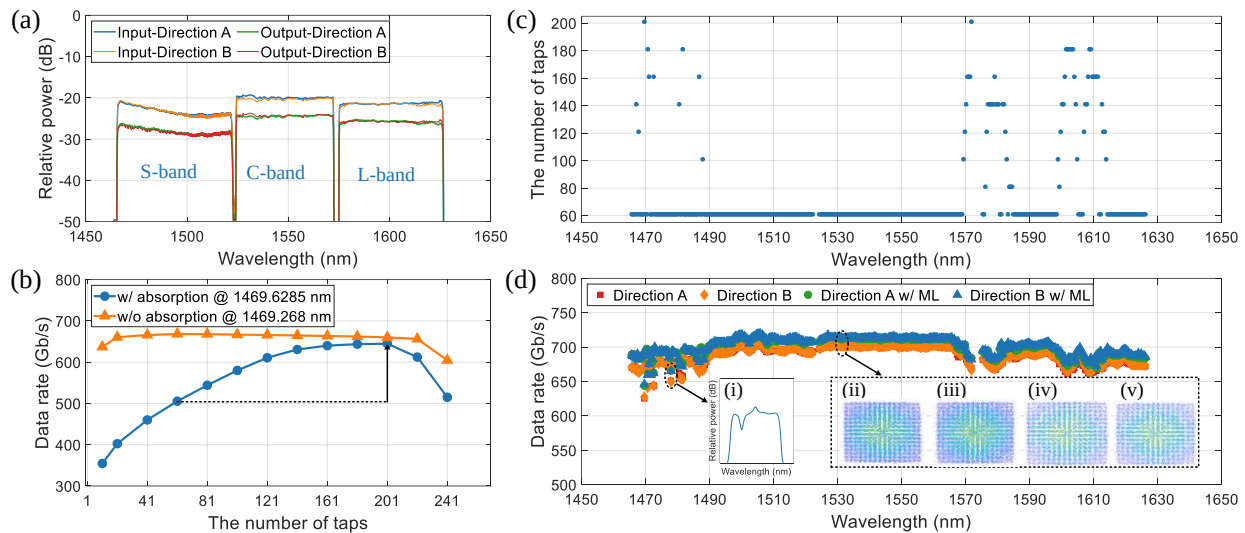


Fig. 2. (a) Optical spectra at the input and output of the NANF. (b) GMI-estimated data rate versus the number of taps, evaluated with and without the effect of water vapor absorption. (c) The number of taps employed for signal equalization across all wavelengths. (d) GMI-estimated net data rate after 10.9-km NANF. Insets (i) Signal spectrum with H₂O absorption at 1477.96 nm; (ii-v) Recovered constellations at 1533.66 nm for Direction A, Direction B, and their counterparts with ML compensation.

4. Conclusions

We have achieved a record net data rate of 550.97 Tb/s by exploiting five dimensions (amplitude, phase, polarization, wavelength, and direction) to carry information, with S+C+L-band transmission over 10.9-km NANF. To the best of our knowledge, this is the highest capacity reported in any single-core single-mode fiber, highlighting the potential of AR-HCF to support high-capacity and ultra-wideband bidirectional transmission.

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5. References

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